

Specs Calculation Report: Bluetooth audio amp, 2x3W class D

Component Summary

1. Speaker (Load per channel) | Qty: 1 | 4 Ohm Impedance, 3W RMS Power Handling
2. Power Supply Voltage (Vcc) | Qty: 1 | 5V
3. Total Average Supply Current (I_supply_avg) | Qty: 1 | 1.33A
4. Output Inductor (L_out) per channel | Qty: 1 | 10uH, Current rating >1.23A (e.g., 2A saturation current)
5. Output Capacitor (C_out) per channel | Qty: 1 | 470nF, Voltage rating >5V (e.g., 10V)
6. Power Supply Bulk Decoupling Capacitor (C_bulk) | Qty: 1 | 470uF, Voltage rating 10V

Detailed Calculations & Reasoning

1. Speaker (Load per channel)

Calc: $P_{RMS} = 3W$, $R_{load} = 4 \text{ Ohm}$ (given) = 4 Ohm, 3W

Reasoning: The amplifier is specified to deliver 3W per channel. 4 Ohm is a common speaker impedance.

2. Power Supply Voltage (Vcc)

Calc: $V_{RMS_load} = \sqrt{P_{out} * R_{load}}$; $V_{peak_load} = V_{RMS_load} * \sqrt{2}$; $V_{cc} \geq V_{peak_load}$

Reasoning: To deliver 3W into a 4 Ohm speaker using a bridge-tied load (BTL) Class-D amplifier, the supply voltage must be at least the peak load voltage.

3. Total Average Supply Current (I_supply_avg)

Calc: $P_{out_total} = 2 * P_{out_per_channel}$; $P_{in_total} = P_{out_total} / \text{Efficiency}$; $I_{supply_avg} = P_{in_total} / V_{cc}$

Reasoning: This is the average current the 5V power supply must deliver to both channels of the amplifier.

4. Output Inductor (L_out) per channel

Calc: $I_{peak_load} = V_{peak_load} / R_{load}$; L_out value is chosen based on typical application for f_sw

Reasoning: Class-D amplifiers require an LC low-pass filter at their output to reconstruct the analog signal.

5. Output Capacitor (C_out) per channel

Calc: C_out value is chosen based on typical application for f_sw and R_load; $V_{rated} \geq V_{cc}$. $f_{cutoff} = 1 / (2 * \pi * L_{out} * C_{out})$

Reasoning: This capacitor forms the low-pass filter with the output inductor. Its capacitance value is chosen to ensure the cutoff frequency is well above the audio band.

6. Power Supply Bulk Decoupling Capacitor (C_bulk)

Calc: C_bulk value is chosen based on I_supply_avg and Vcc ripple requirements (typical value); $V_{ripple} = I_{supply_avg} * \Delta t / C_{bulk}$

Reasoning: This electrolytic capacitor provides localized energy storage for the amplifier, helping to maintain a steady supply voltage during transient loads.